AI Agent Assignment

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Class: BS 6A

Section: A

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Agent Type: Food Delivery Agent

1. Identify the Problem

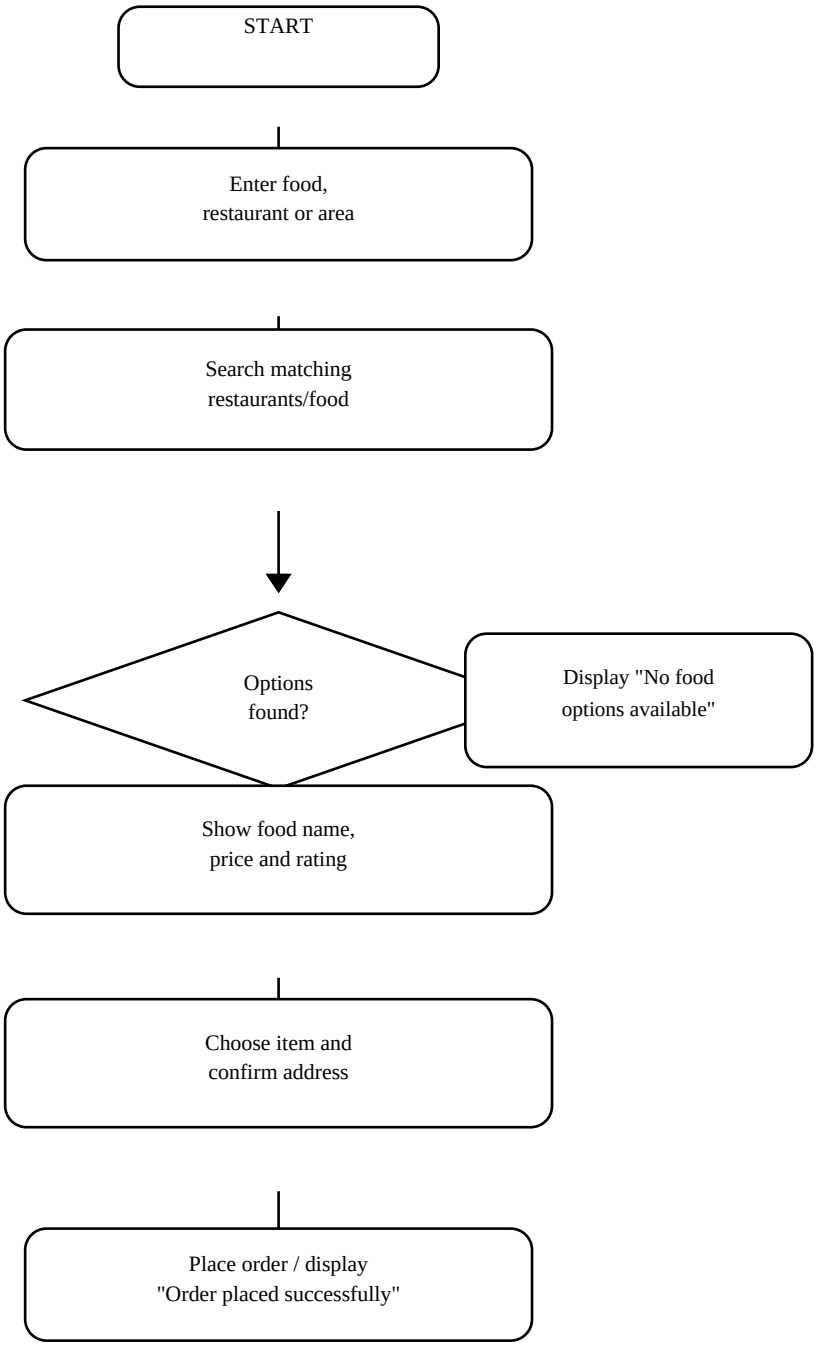
Usually, when someone wants to order food, they have to search through different restaurants, check menus, compare prices, and contact the restaurant or delivery service. This can take a lot of time and make it difficult to choose a suitable option. To solve this problem, we have designed a Food Delivery Agent that takes the user's preferred food or restaurant as input, searches for matching options, shows the available choices along with their prices and ratings, and places the order easily once the user makes a choice.

2. Formulation

2.1 Textual Formulation

```
START
Ask the user to enter preferred food (or restaurant/area)
Search for restaurants and food items that match the request
IF no matching food/restaurants are found THEN
   Display "No food options available"
STOP
ELSE
   Display list of available food options with name, price, and rating
   Ask user to choose a food item
   Confirm delivery address and place the order
   Display "Order placed successfully"
END IF
END
```

2.2 Graphical Formulation (Flowchart)


START
Enter food,
restaurant or area
Search matching
restaurants/food
Options
found?
Display "No food
options available"
Show food name,
price and rating
Choose item and
confirm address
Place order / display
"Order placed successfully"

3. Evaluation in the Context of PEAS

Element	Description
Performance Measure	How accurately the agent matches food and restaurants to the user's demand, whether the order is placed correctly, whether the delivery details are confirmed properly, and how fast it responds.
Environment	The user, restaurant and menu data, food prices and ratings, delivery locations, and available delivery options.
Actuators	Showing the list of matching food options on screen, displaying prices and ratings, confirming the selected item, and displaying the order confirmation.
Sensors	The user's input (preferred food, restaurant, or area), the selected food item, and the user's delivery address.

4. Environmental Properties

Property	Explanation
Partially Observable	The agent only knows what the user tells it (the food, restaurant, area, and delivery address); it does not have a direct view of the whole restaurant or delivery environment.
Deterministic	If the user enters the same food or restaurant request under the same available data, the agent will return the same matching options.
Sequential	All the steps happen in a particular order: first search, then show the results, then choose an item, confirm delivery details, and place the order. Earlier actions affect later steps.
Static	The food and restaurant information is treated as unchanged while the agent is searching or waiting for the user's response.
Discrete	Food items, restaurant names, prices, ratings, and delivery choices come from a fixed, countable set rather than a continuous range.
Single-agent	Only this one food delivery agent is working in this environment; there is no other competing agent.